Data Flow Diagrams (DFD) & Decision Specifications

Academic Grievance, No-Dues & Clearance Automation Engine

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Course: Software Engineering
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Document Scope: Level 0 (Context), Level 1 (Decomposition), Core Decision Table, Core Decision Tree, and Detailed Process Explanations

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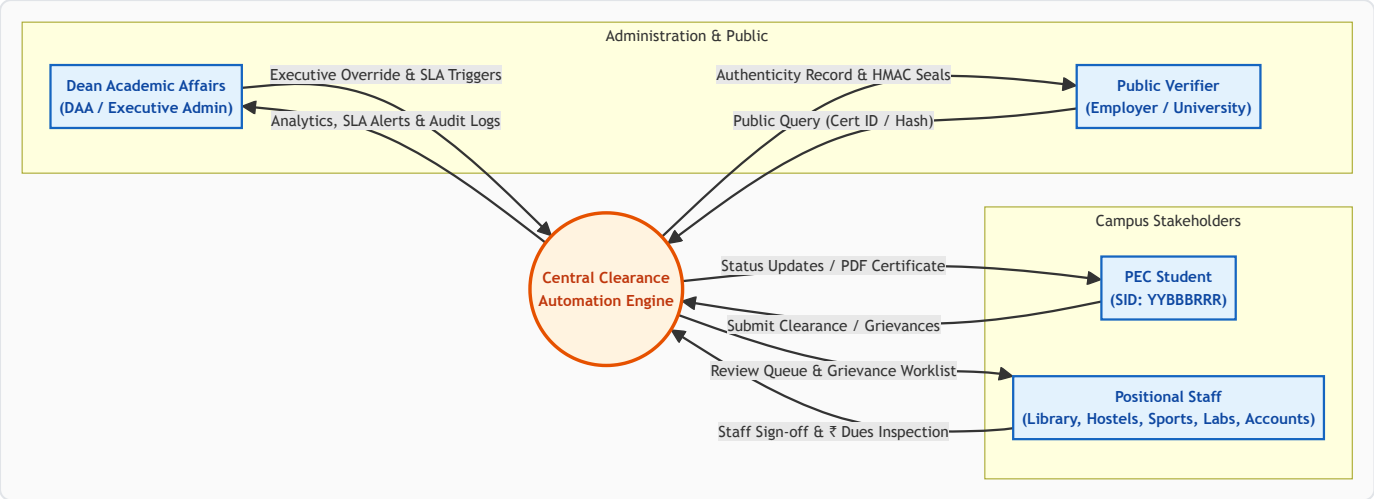
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1. DFD Level 0: Context Analysis Diagram

The **Level 0 Context Diagram** defines the primary environmental boundary of the PEC Clearance Automation Engine, mapping all interactions between campus stakeholders and the central automated processing engine.

Level 0 Context Diagram



Architectural Explanation & External Entities

The Context Diagram encapsulates the entire institutional software system within a single high-level transformation bubble, isolating internal operations from four external entities:

1. **PEC Student (`Student`):**
   - **Identity:** Identified by an immutable 8-digit PEC Student Identifier (`YYBBBRRR`, e.g., `22103042`).
   - **Function:** Applies for clearance at degree completion or milestone exit, monitors real-time progress across all 5 departmental facilities, reviews itemized outstanding liabilities in Indian Rupees (`₹`), files grievance appeals against disputed charges, and retrieves digitally signed vector PDF certificates.
2. **Positional Department Staff (`Staff`):**
   - **Identity:** Positional signatories representing Dr. B.R. Ambedkar Central Library, PEC Hostels, Gymkhana (Sports), Department Computing/Hardware Labs, and the Accounts Branch.
   - **Function:** Authenticates under facility role scopes, inspects physical asset returns, records itemized liabilities (`due_amount` in INR), issues digital approvals sealed with keyed HMAC-SHA256 tokens, and arbitrates student grievance tickets.
3. **Dean Academic Affairs (`Admin` / DAA):**
   - **Identity:** Executive academic oversight persona.
   - **Function:** Monitors campus-wide clearance efficiency and department bottleneck latency, receives high-priority alerts for breached Service Level Agreements (SLAs), and possesses executive authority to override clearance rejections with mandatory audit justification.
4. **Public Certificate Verifier (`Verifier`):**
   - **Identity:** External employers, background screening agencies, and international universities.
   - **Function:** Performs tokenless, zero-login cryptographic verification of issued certificates using either the unique Certificate ID (`PEC-CLR-YYYY-XXXXX`) or the 64-character SHA-256 public hash digest.

Boundary Data Flows Summary

- **Inbound Data Flows:**
  - `Clearance Application`: Student SID, academic year, semester, and clearance justification.
  - `Grievance Submission`: Department target, subject, dispute narrative, and priority tier (`LOW`, `MEDIUM`, `HIGH`, `CRITICAL`).

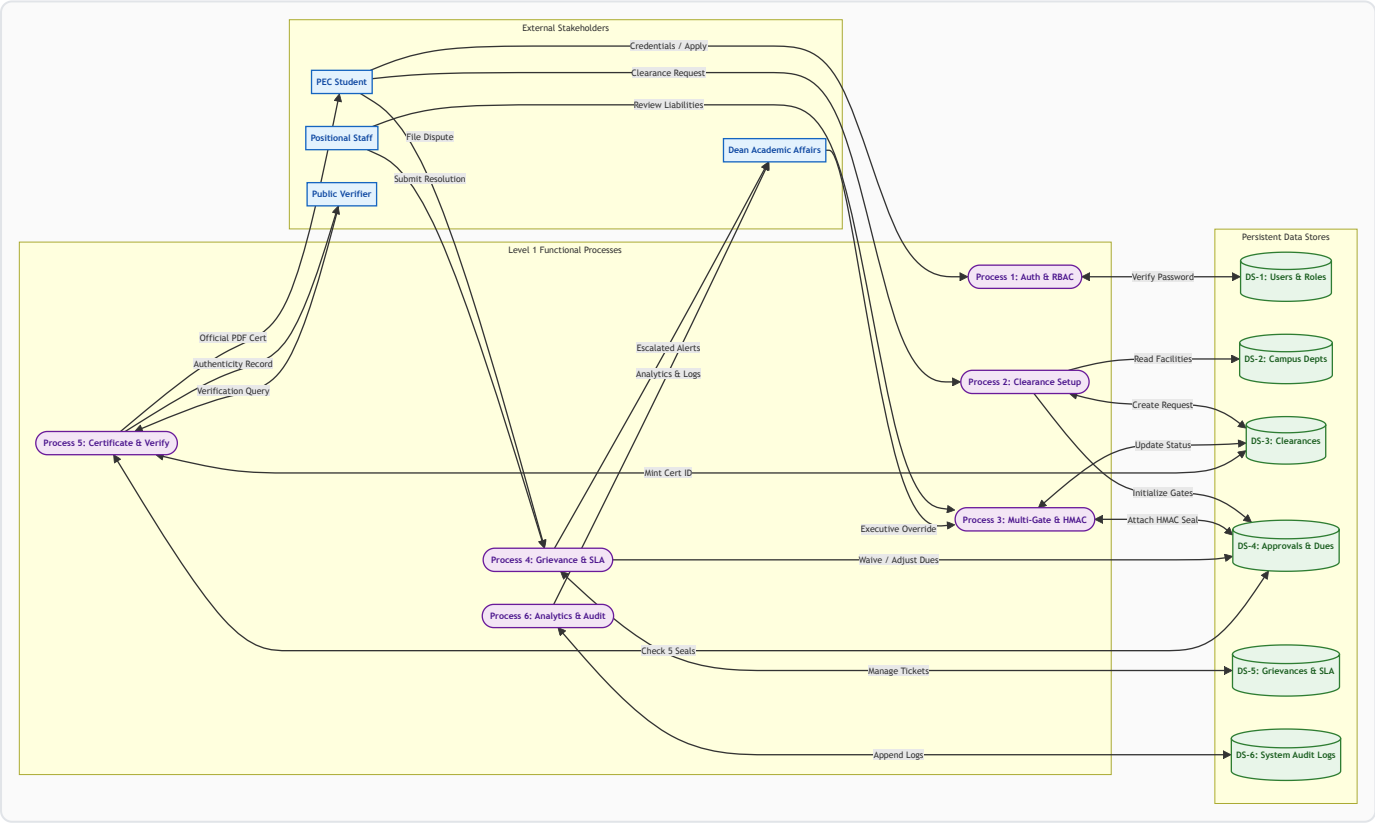
    - `Liability Inspection & Sign-off`: Approval decision (`APPROVED` or `REJECTED`), itemized dues in INR (`₹`), and reviewer commentary.
    - `Executive Override Order`: Clearance ID, department code, and mandatory DAA justification.
    - `Public Verification Query`: Certificate ID string or SHA-256 hash digest.
- **Outbound Data Flows:**
    - `Live Pipeline Status`: Multi-gate progress status and itemized dues across all facilities.
    - `Tamper-Proof PDF Certificate`: Printable vector PDF embedded with institutional seals and SHA-256 digest.
    - `Department Review Queue`: Filtered list of pending student clearance records restricted to staff's facility.
    - `SLA Breach Alerts`: Real-time administrative notifications for overdue grievances.
    - `Authenticity Verification Record`: JSON payload confirming validity, student metadata, and 5 departmental HMAC seals.

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2. DFD Level 1: System Decomposition Diagram

The **Level 1 System Data Flow Diagram** decomposes the top-level system into six dedicated, single-responsibility operational subsystems (`Process 1` through `Process 6`), interacting directly with six persistent relational data stores (`DS-1` through `DS-6`).

Level 1 System Data Flow Diagram



Functional Subsystems Explanation (Processes 1 to 6)

1. **Process 1: User Authentication & Role Authorization (`auth.py`, `security.py`)**
   - **Function:** Authenticates inbound users against Bcrypt-salted password hashes, validates PEC institutional domains (`@pec.edu.in`), issues signed cryptographic JSON Web Tokens (JWT), and checks role claims (`STUDENT`, `STAFF`, `ADMIN`) to protect API routes.
   - **Inputs:** Raw email and plaintext password from Student, Staff, or Admin; stored hashes from `DS-1`.
   - **Outputs:** Signed Bearer JWT access tokens with encoded identity claims (`sub`, `role`, `exp`).
2. **Process 2: Clearance Application & Gating Setup (`clearance.py`)**
   - **Function:** Enforces the institutional single-active-request invariant, captures student degree milestone details, provisions the master clearance record in `DS-3`, and initializes 5 departmental approval records in `DS-4` (Gate 1 facilities set to `PENDING`; Gates 2 & 3 set to `WAITING_PREREQUISITE`).
   - **Inputs:** Clearance application packet from Student; active facility records from `DS-2`.
   - **Outputs:** Master clearance record in `DS-3`; 5 initialized approval rows in `DS-4`; audit event to `DS-6`.
3. **Process 3: Departmental Review, Chain of Responsibility & HMAC Signing (`chain_of_responsibility.py`)**
   - **Function:** Orchestrates the multi-signatory clearance pipeline using the behavioral Chain of Responsibility design pattern. Enforces parallel reviews at Gate 1, sequential dependency barriers at Gate 2 and Gate 3, calculates keyed HMAC-SHA256 digital seals upon approval, handles DAA executive overrides, and mutates master clearance status.
   - **Inputs:** Staff decisions (`APPROVED` / `REJECTED`), itemized dues in INR (`₹`), DAA override commands; current approval states from `DS-4`.
   - **Outputs:** Cryptographic HMAC signature seals attached to `DS-4`; unlocked downstream gates; master status update in `DS-3`.
4. **Process 4: Grievance Redressal & SLA Escalation Engine (`grievance.py`)**
   - **Function:** Provides an institutional dispute mechanism for students contesting incorrect dues or missing records. Automatically assigns dynamic SLA deadlines based on ticket priority (`CRITICAL`: 4h, `HIGH`: 12h, `MEDIUM`: 24h, `LOW`: 48h). Scans for expired tickets and triggers automated escalation banners directly to the DAA dashboard.
   - **Inputs:** Student grievance filings; staff resolution decisions; periodic SLA scan pulses.
   - **Outputs:** Grievance records in `DS-5`; adjusted/waived dues in `DS-4`; real-time SLA breach alerts to Admin.
5. **Process 5: Certificate Minting, PDF Rendering & Verification (`certificate.py`, `pdf_generator.py`)**
   - **Function:** Validates that all 5 departmental facilities have recorded `APPROVED` statuses with valid cryptographic HMAC seals. Unlocks the release of the student's ₹10,000 Caution Money refund, mints a permanent Certificate ID (`PEC-CLR-YYYY-XXXXX`), calculates a 64-character SHA-256 verification hash, generates an official vector PDF with letterhead, and serves tokenless public verifications.
   - **Inputs:** 5 valid approval records from `DS-4`; clearance record from `DS-3`; public verification queries.
   - **Outputs:** Minted certificate details in `DS-3`; printable binary PDF to Student; verification JSON to Verifier.
6. **Process 6: Institutional Administration, Analytics & Audit (`admin.py`)**
   - **Function:** Computes institutional clearance metrics (overall completion rates, average department review latency, clearance bottlenecks), and maintains an immutable append-only system audit log tracking every security and operational event.
   - **Inputs:** Internal lifecycle event signals; aggregate queries across `DS-3`, `DS-4`, and `DS-5`.
   - **Outputs:** Real-time campus analytics payloads and append-only audit entries in `DS-6`.

Persistent Data Stores Explanation (DS-1 to DS-6)

- **`DS-1: Users & Personas` (`users` table):** Stores user credentials, salted Bcrypt password hashes, personas (`STUDENT`, `STAFF`, `ADMIN`), student roll numbers (SIDs), and staff department assignments.
- **`DS-2: Campus Departments` (`departments` table):** Institutional configuration of the 5 facilities (Library, Hostels, Sports, Labs, Accounts), gate classifications (Gate 1, 2, or 3), and positional head titles.
- **`DS-3: Clearance Requests` (`clearance_requests` table):** Master clearance records storing student references, academic session, overall status (`PENDING`, `APPROVED`, `REJECTED`), certificate ID, and public SHA-256 hash digest.
- **`DS-4: Department Approvals & HMAC Seals` (`department_approvals` table):** Operational approval rows containing review status (`WAITING_PREREQUISITE`, `PENDING`, `APPROVED`, `REJECTED`), itemized dues in INR (`due_amount`), reviewer remarks, and keyed HMAC-SHA256 signature seals.
- **`DS-5: Grievances & SLA Timers` (`grievances` table):** Dispute tickets holding priority classification, computed absolute SLA expiration timestamps, escalation flags (`is_escalated`), staff responses, and resolution timestamps.
- **`DS-6: System Audit Logs` (`audit_logs` table):** Append-only audit trail capturing actor ID, actor role, immutable action verb, affected entity ID, textual details, and microsecond UTC timestamps.

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3. Clearance Pipeline Decision Logic: Table & Tree

The core algorithmic innovation of the engine is the **Multi-Gate Clearance Pipeline & Barrier Enforcement Logic**, implemented via the behavioral **Chain of Responsibility** pattern. Below are the definitive Decision Table and Decision Tree governing all pipeline transitions.

Decision Table: Multi-Gate Clearance Pipeline Matrix

This decision table formalizes the state transition and gating barrier rules enforced by the **Chain of Responsibility** engine (`Process 3`):

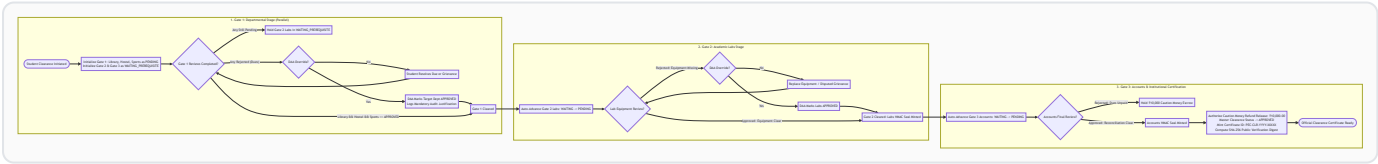
Rule ID	Gate 1 Status (Lib / Host / Sport)	Gate 2 Status (Dept Labs)	Gate 3 Status (Accounts)	DAA Override?	Action Target & Enforced State Transition
R-101	All `PENDING`	`WAITING`	`WAITING`	No	Allow concurrent Gate 1 reviews; mutate to `APPROVED` or `REJECTED`; mint HMAC seal on approval.
R-102	Any `PENDING` / `REJECTED`	`WAITING`	`WAITING`	No	**BLOCK GATE 2 WITH HTTP 400:** Reject review attempt; Gate 1 facilities incomplete or disputed.
R-103	Any `REJECTED` (Dues > ₹0)	`WAITING`	`WAITING`	No	**REJECT MASTER STATUS:** Set overall request to `REJECTED`; prompt student to settle dues or file grievance.
R-104	All `APPROVED`	`WAITING`	`WAITING`	No	**AUTO-UNLOCK GATE 2:** Automatically advance Gate 2 (Dept Labs) from `WAITING` to `PENDING`.
R-105	All `APPROVED`	`PENDING`	`WAITING`	No	Allow HOD Labs review; inspect lab kits; if clear, mint HMAC seal and set status to `APPROVED`.
R-106	All `APPROVED`	`APPROVED`	`WAITING`	No	**AUTO-UNLOCK GATE 3:** Automatically advance Gate 3 (Accounts) from `WAITING` to `PENDING`.
R-107	All `APPROVED`	`APPROVED`	`PENDING`	No	Allow Accounts review; reconcile dues; authorize release of ₹10,000 Caution Money deposit refund.

Rule ID	Gate 1 Status (Lib / Host / Sport)	Gate 2 Status (Dept Labs)	Gate 3 Status (Accounts)	DAA Override?	Action Target & Enforced State Transition
R-108	All `APPROVED`	`APPROVED`	`APPROVED`	No	**MASTER CLEARANCE GRANTED:** Mutate master status to `APPROVED`; mint Certificate ID & SHA-256 Digest.
R-109	Any `REJECTED`	Any	Any	**Yes (DAA)**	**EXECUTIVE OVERRIDE:** Mark target department `APPROVED`; clear dues; log `ADMIN_OVERRIDE` in `DS-6`.

Decision Table Explanation (Chain of Responsibility & Barriers)

1. **Gate 1 Parallel Independence (Rule R-101):** The three Gate 1 departments (Central Library, Hostels, Sports) operate concurrently. A rejection or delay in one does not block staff members in the other two from inspecting assets or recording sign-offs.
2. **Sequential Dependency Barriers (Rules R-102 & R-106):** Gate 2 (Department Labs) cannot be reviewed until all three Gate 1 facilities are `APPROVED`. Any attempt by lab staff to sign off prematurely is rejected with HTTP `400 Bad Request`. Similarly, Gate 3 (Accounts) cannot be accessed until Gate 2 is fully cleared.
3. **Automated Pipeline Advancement (Rules R-104 & R-106):** The system dynamically scans prerequisites upon every sign-off. When the final Gate 1 facility is approved, Gate 2 automatically transitions from `WAITING_PREREQUISITE` to `PENDING`. When Gate 2 is approved, Gate 3 automatically unlocks.
4. **Caution Money Disbursement & Master Certification (Rules R-107 & R-108):** Accounts Branch review triggers financial reconciliation. Once Gate 3 is approved, the institutional Caution Money escrow deposit (₹10,000.00) is marked for release, and the master request status becomes `APPROVED`.
5. **Executive Override Powers (Rule R-109):** The Dean Academic Affairs retains statutory authority to override departmental rejections (e.g., in cases of administrative error or approved waivers), immediately setting the target department to `APPROVED` and recording a mandatory audit trail entry in `DS-6`.

Decision Tree: Clearance Gating & Barrier Enforcement



Decision Tree Explanation

The Decision Tree visually traces the deterministic multi-gate clearance lifecycle:

- **Phase 1 (Gate 1 Execution):** Initializes the three non-academic facilities concurrently. If any liability is found, the clearance halts with status `REJECTED`, prompting the student to pay or file a grievance. Only when all three facilities record `APPROVED` (or receive DAA override) does the pipeline advance.
- **Phase 2 (Gate 2 Execution):** The automated engine advances Gate 2 from `WAITING` to `PENDING`. Academic computing/hardware labs verify asset returns. Approval seals the department record with an HMAC-SHA256 signature; rejection blocks further progress.
- **Phase 3 (Gate 3 Execution):** Accounts Branch reconciles tuition, hostel, and fee accounts. Once approved, the ₹10,000 Caution Money refund is authorized, the master status becomes `APPROVED`, and the system generates the unique Certificate ID and public verification digest.